

The ethics of generative AI in software engineering

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1 Introduction

Software engineering is a field that has developed most of today's infrastructure in some way or another, be it via the implementation of a medical device to the websites that allow us to connect the world. However, these projects need to manage several stakeholders, engineers and budgets to make things come to reality. With the improvements in generative AI (GenAI for short), there has been a shift towards trying to incorporate this new tool into the software development pipeline. This tool has been adopted by industry leaders such as Google, Microsoft, Amazon and more. [17, 25, 27].

Given the workings of generative AI, there are several key concerns: The displacement of people for compute centers, the correctness of the code

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generated and the damage to growth spaces for future engineers. This essay will try to answer the question:

“Is it ethical for software engineers to use modern generative AI applications to generate code that is used, in whole or in part, for commercial applications?”

2 Core Issues

As discussed in the introduction, there are several concerns with the use of generative AI in software development:

2.1 Code correctness

One of the major issues that stems from using GenAI is the fact that GenAI doesn't *reason* like humans do given its answer is generated via approximating using other good answers in the source. Programming is a task that requires discrete reasoning as the steps will directly relate to the next one in the solution [6]. This causes several issues when it comes down to the reliability of the software being written with the help of GenAI [15].

According to a 2024 study conducted by the International University of Applied Sciences, the amount of code generated is about 97% of the times executable. While this is a high rate, the issues start to stem from the correctness, where about 21% of the generated lines of code were fully correct and about 18% were able to be corrected minimally to achieve the desired result [15]. While the methods of the study focused on smaller programs with simpler functionality, the risks of failure can be

extrapolated to larger systems where the codebase is larger and more side effects need to be taken into account when submitting changes.

A production instance of code correctness not being up to par is Amazon's outages in 2026. The outage of major Amazon services was reported to be caused by a GenAI tool used for code. Repeated instances of this are being prevented by strengthening the guard rails present for change submissions, specially those that used GenAI in their development. However, it is important to highlight major services like Amazon Web Services (AWS) went down due to GenAI changes [20]. The issue here is not AWS being down but instead the clients of the service being collateral damage of such infrastructure failure without a clear accountability chain.

As seen with AWS, substantial damage can be done to infrastructure even when there are senior engineers in the production process. The risks become more dangerous with issues like GenAI hallucination [14] or the use of GenAI code in devices and systems that require critical safety such as biotechnology, embedded systems in transportation and more. In those cases, a model converging to the wrong answer can cost lives and without a clear accountability chain, it is difficult to enforce justice and repair damages. There are several issues with deciding how to hold accountability as mentioned in an article by IBM: "With the C-suite? With the whole team? If you have accountability that's spread over the entire organization, everyone can't end up in jail. Ultimately, shared accountability often leads to no accountability." [9].

2.2 Damage to growth spaces for entry level engineers

Another major issue with GenAI for software engineering is the fact that they reduce the spaces where less experienced engineers build the knowledge to move forward. As stated before, GenAI doesn't reason and this is a similar issue with less experienced engineers, however, the key difference is that human engineers can improve with time and build up the discrete reasoning which

allows them to construct correct programs that they haven't seen before. [21]

As mentioned in an article by computer world, it is mentioned how Amazon cut around 14000 jobs because of the developments in GenAI technology allowing them to replace the humans with machines [25]. This presents a major issue as there is still a need for engineers in the production pipeline, even more so when mistakes like the outages necessitate more human reviews of the submitted code [20]. This pattern is industry wide as there is a 25% decrease in entry level tech job hiring and a decrease of about 30% in internship positions [21].

In an blog post by Stack Overflow, the author discusses the importance of hiring entry level engineers as this facet of the tech industry is apprenticeship heavy given the complex systems that are being produced and maintained that are simply not available in an academic context. The article also concludes with the call to action to senior engineers and managers highlighting how there will be no other way to train entry level engineers beyond hiring them [18].

2.3 Displacement of people

This is a more general issue with the usage of GenAI but applies heavily to expensive tasks like generating the code for a complex program. The infrastructure that GenAI needs is costly in several resources [19, 28]:

- About 22% of all electricity used by US households per year.
- "Chan School of Public Health found that the carbon intensity of electricity used by data centers was 48% higher than the US average."
- "Large data centers can consume up to 5 million gallons of water per day, equivalent to the water use of a town populated by 10,000 to 50,000 people."

In addition to the resource costs, data centers require space to be built. In the past year, there have been about 831 data centers built across the world to meet the GenAI demand. These sites have been constructed across the world [7]. While this

will decrease resources available for communities where the centers are built, the data centers will affect communities in developing countries the most. This is due to companies concealing their identities to purchase the land, and in some cases, not needing to disclose the resources they will be consuming, such as Microsoft's license to exploit a well that has been overexploited in Mexico [5].

3 Stakeholders

3.1 General Population

The general population is relevant to the usage of GenAI in software development as they are the customers of the products being made with GenAI support. Not only that but they are also the people getting affected most when a new AI compute center.

3.2 Software engineers

Software engineers are relevant to the situation as they are the people being encouraged to use the tool and being displaced by the implementation of the tool in the software production pipeline [17, 21]. This places software engineers in an interesting position as they might be the ones taking accountability for the mistakes of AI and future infrastructure suffers because of the displacement of entry level developers [9, 18].

3.3 Companies that use generative AI

Companies that use GenAI are using a tool that has major risks such as Amazon's outages [20]. This creates a conflict of interest, in cases like Google where they are the company developing the tool as well as selling it [17]. Additionally, for the client companies of the GenAI vendors, there is the documented issues from using GenAI in production [12] as well as the lack of transparency of which products were made using GenAI for code generation (e.g. there is no specific reporting of the amount of AI generated code in products).

3.4 Companies that make generative AI tools

These companies benefit monetarily from the use and perceived success of GenAI tooling such as Claude Code, Google Gemini, etc. These companies are also responsible for transparency in the origin of the data used, the errors their tool might make and the data it will collect from the project [4, 20, 27].

4 Current state of generative AI

4.1 Usage in research

There's been a shift in academia to use more GenAI to speed up different elements of the research pipeline [1]. A key project that will be mentioned in this essay is the usage to create the implementation for the ZOZO's (a Japanese clothing company) cloth physics solver [2, 3].

4.2 Usage in open source

In open source projects there is a current issue where the lead contributors are being swarmed by pull requests with low quality AI generated code [16]. This puts a strain in projects' ability to survive as the leads are often people who volunteer and don't have the time to go through all the low quality code to pick out and accept contributions to the project.

4.3 Usage in private companies

Across different companies, there are several philosophies, some companies are encouraging their developers to use GenAI and needing developers to strengthen the code review process [8, 25] and other companies avoid using GenAI as they believe it damages the quality of their products [13].

5 Technical Background

5.1 Machine learning

Machine learning is a subset of artificial intelligence that uses statistical inference to adapt responses based on previous and new knowledge. This is a very powerful type of program as it provides adaptability at no code architecture change, however, these systems improve drastically the more data it has available and analyzed [10].

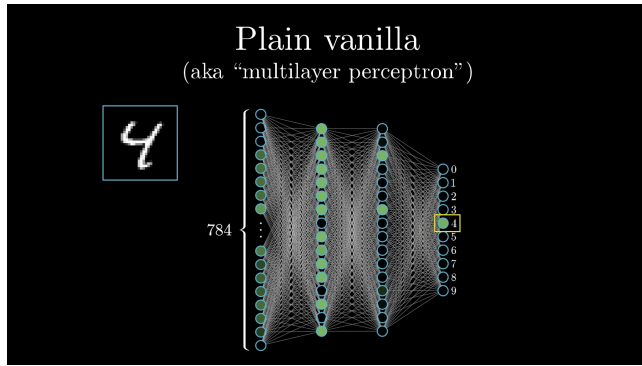


Figure 1: Neural network made to recognize numbers 0 to 9 on a 28 by 28 grid [22].

Models work via the implementation of neural networks. A neural network is composed of *neurons*, which are simply just a number that represents the presence of a component in the data (the circles in the image 1). A vector of neurons represents a layer, this is used to represent different components of the data. Neural networks are normally composed of several layers each neuron connected to another via weights (the lines in image 1). This allows the model to encode the activation of a neuron based on a linear combination of the weights and neurons of the previous layer. In the image 1 the activation is represented by how green each circle is. Each layer encodes different sets of properties about the data, in the case of figure 1 the first layer represents how bright each pixel is while the next one represents relationships in the distributions. The first layer is one of the few layers that can be explicitly linked to the data as the activation is entirely determined by the model maker. This is because these linear combinations

can be fine tuned with a bias term to ensure that the model produces better results [22].

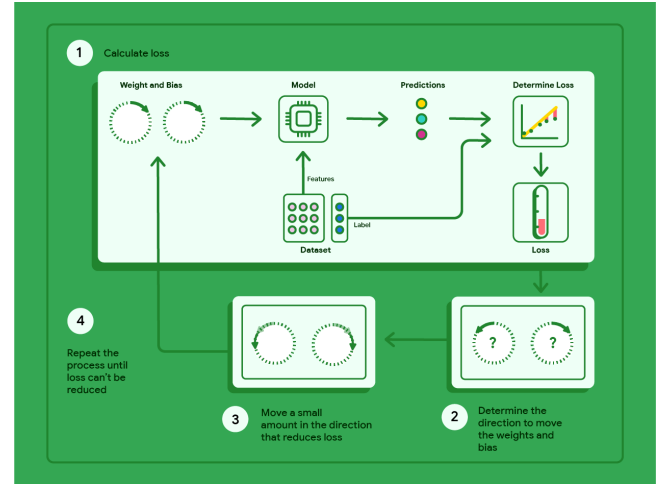


Figure 2: The model training pipeline [11]

The training process for a model involves utilizing data that is labelled to be the right response and iteratively improving all of the model's weights and biases (denoted by $\vec{w}$). This process of improvement utilizes cost function and gradient descent. The cost function (ϵ) describes the error of the current model based on the given parameters, this error is normally averaged to improve across all input types instead of only one [11]. Gradient descent is an iterative optimization method defined as such:

$$\vec{w}' = \vec{w} + \gamma \begin{bmatrix} \frac{\partial \epsilon}{\partial w_1} \\ \vdots \\ \frac{\partial \epsilon}{\partial w_n} \end{bmatrix}$$

This method is used for steps 2 and 3 in figure 2 as it will calculate the direction to move the weights and biases and move there. This process is repeated until the change in error between iterations is less than desired [11]. While this only adjusts the weights, there still needs to be a process of back-propagation. This involves finding which parameters affect the output the most and then

adjusting the values to cover the edge cases left after the gradient descent is done ¹ [23].

5.2 Large Language Models

Large Language Models (LLMs) are models that are trained on natural and computer language in order to be able to generate an output based on verbal (text and/or audio) input. These models use the relationships encoded with machine learning to output next most likely word [24].

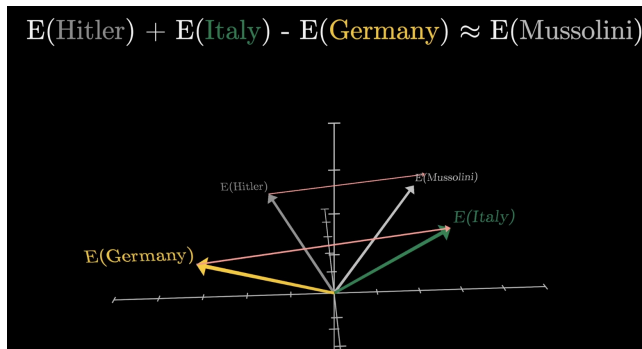


Figure 3: Visuals of how the embedding matrices transform different words into different "word spaces"[24]

When the user submits a prompt, the model will turn the prompt into tokens. Tokens are small chunks of the prompt that can be turned into vector values (each word being like a pixel in figure 1). This is done with an embedding matrix, which describes the conversions between a token and their respective vector value. This vector value, like discussed in the machine learning section, represents a set of relationships be it about the word itself or the semantic context of the word [24].

After that, each vector will affect each other to ensure that the context is taken into account, this is known as the attention block (see figure 4) [24].

After finalizing the contextualized vectors, the model will then pass each vector separately through the layers of the neural network. This will result with a final vector like the one in figure 1, but the

¹The math of this algorithm is more complex therefore it will not be explored in depth for this essay

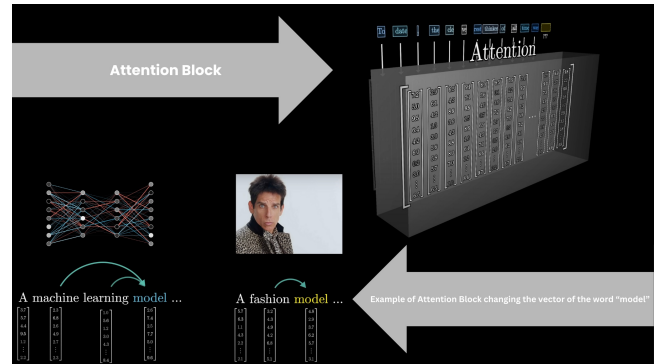


Figure 4: Visuals of how the attention block uses and represents context numerically [24]

vector will represent a scalar of how relevant the word is.

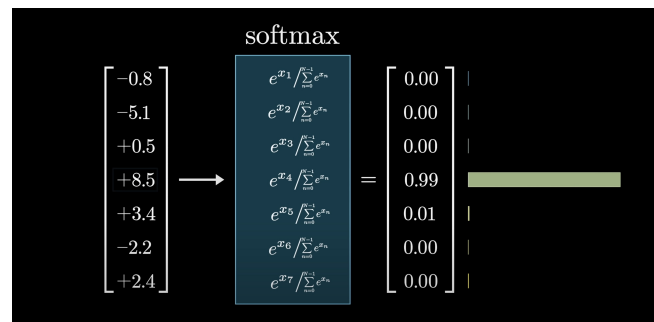


Figure 5: How the scalar values are turned into a probability distribution [24]

As the scalars are not a probability, the procedure in figure 5 is performed. This then allows the model to pick either the most likely word with a bit of added randomization to mimic natural language [24].

This process is conducted iteratively until the response is fully formed. The nice thing about this model is how it can turn the set of input tokens into a set of different parameters used to construct an output. While this example focused on generating text, this extends to code with the main difference being that the output vector is not composed of words or punctuation but symbols and language tokens (such as brackets or keywords) [26].

5.3 Architecture of GenAI coding tools

When utilizing generative AI in the context of a software project, there are additional considerations that must be met to meet the desired output. To better understand this fine tuning, this essay will focus on Claude Code's approach as a case study.

According to the Claude Code documentation: "Claude Code is an agentic coding tool that reads your codebase, edits files, runs commands, and integrates with your development tools. Available in your terminal, IDE, desktop app, and browser."

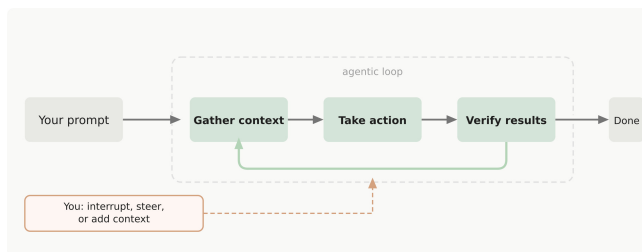


Figure 6: Claude Code's prompt processing architecture [4]

When a developer prompts Claude Code, it will read all the files it is given access to (such as the git index, the other source files in the directory, etc.) for relevant symbols and information related to the prompt (Think about the attention block discussed in the previous sections). Then it will generate the code snippet and/or commands to run in the terminal to execute the prompt. This is done using process discussed in the previous section. Once the prompt output is generated, it is verified by compiling the program and running it and verifying the output matches the desired results [4]. This entire process can be interrupted by the developer as seen in figure 6.

6 Ethical Analysis

Given the broadness of the issue presented in the essay, this section will be split into the core issues presented in section of the same name.

6.1 Code Correctness

As discussed in the technical section, AI utilizes the relationship between the provided prompt and the relationships it can build with the data to generate the code snippet. There is a major issue in this generation process, it doesn't perform logical reasoning in order to ensure that the code is correct. Tools like Claude utilize the added knowledge of the database, compiler errors and program output to ensure the program performs what is desired. However, this poses a critical issue, there is no guarantee that the side effects that stem from the lines being executed (such as memory allocation, pointer dereferencing, etc) don't harm the project or the users. This can be seen in several cases where the models are not trained with code of the quality and standards desired, which in turn leads to bugs [12].

According to an article by Computer World, Amazon recently fired 14000 employees because of AI being able to to perform their jobs at higher speeds instead of hiring more personnel [25]. However, this presents ethical concerns under Kantianism. When companies use GenAI as a cost reduction method, they are using people as means to an end. First, the training data used to train the model was not obtained with explicit permission. Second, it disregards the customer's humanity by selling them a product that is more likely to have failures and not being transparent about the components made with GenAI.

In the context of act utilitarianism, the major costs involve the damages that a malformed product can cause and the benefits are the time and costs saved in business. When the software written with GenAI deploys, there are millions of people relying on it. A key example is the Amazon outage in early 2026, where the sellers and consumers get negatively impacted by the infrastructure crashing due to a GenAI induced bug and the engineers not catching it in review [20]. On the other hand, the saving of costs allowing companies like Google to generate more than half the code with GenAI [17] which allows the engineers and c-suite to make

more money in the long term. When the infrastructure works and the cost is saved, only a select group of people benefit, however, this comes with the risk of harming millions in the process when a failure occurs.

From these two frameworks it can be seen that there are major issues that must be solved before GenAI can provide benefits to developers and companies that will outweigh the costs to the consumer. Given the irreversibility of the data trained, Kantianism will always weigh against GenAI used in production. However, if the benefits can outweigh the cost as the technology improves, it might become more ethically sound to rely on this technology.

6.2 Damage to growth spaces for entry level engineers

When discussing the usage of GenAI in software development, there is a core issue that can't be seen today, the lack of training for younger engineers. A remarkable example is the clothing simulation contact solver paper written for ZOZO and published in the ACM journal. This paper also has an implementation of the technique discussed in the paper on GitHub [2, 3].

Under Kantianism, the development of the project is using people as means to an end as the code generated for the project was trained (the models being MS Copilot and Claude Code) with stolen data. In many universities and companies, there are entry level research positions, this is with the purpose of strengthening newer developers so there are more researchers in the future. With the usage of GenAI in a project like this, it removes the opportunity for a developer to acquire the skills in practice so that eventually they can innovate and research independently. This causes a contradiction as the researchers right now would not be where they are had they not been able to build such experience.

Under act utilitarianism, this issue arises in another context, the open source contribution space.

In open source, most of the code written is by volunteers but used by industry with notable open source projects being Linux, Yocto, Chromium and Android. In the current job market, a way for entry level engineers to gather experience is to contribute to these projects. However, with the flood of GenAI contributions, lead contributors are being stretched thin with low quality code [16]. This is something that damages open source as it develops at a slower pace and in turn damages the whole industry. The cost of GenAI being used for code generation outweighs any of the benefits it has in the open source and commercial space when it comes to the opportunities for developers to grow.

Lastly, virtue ethics argues against using GenAI because an individual who chooses to go with GenAI is actively removing the chance of an entry developer to understand and cooperate with them in the future [18]. This goes against the virtues of cooperation and sustainability.

6.3 Displacement of people

As discussed before, there are several costs to the usage of GenAI in software development, however, this is mainly discussed in the context of the industry. While focusing in the industry is required to be able to find the localized impacts, it is important to consider how the general public is also being disproportionately affected because of the infrastructure needed for this technology to exist [5].

This is not sound ethically under several frameworks. Under Kantianism, it is taking advantage of small communities in developing nations and damaging the environment of the region like the overexploited water resource in Mexico. Under virtue ethics, the individuals at companies that build and sell products using tools with massive negative environmental impacts are not practicing the virtues of collaboration, sustainability and impartiality. Lastly, under act utilitarianism, the usage of GenAI for commercial gain is sacrificing the long term gains of everyone given the carbon intensity of GenAI compute centers [19].

7 Conclusion

Based on the impacts GenAI has on the tech industry as well as outside, it can be seen that the sustainability of the decisions being made damages the ethical feasibility of the technology. If further developments reduce the damage caused to the general public and the environment it will be more ethically sound but the concerns where growth opportunities and work being stolen will not be fixed unless the training procedures being used by the GenAI engineers changes.

8 Beyond the superficial

The question of this essay is very relevant to the course as it is a very new technology with multiple facets to analyze. This is important because while now it is GenAI for code generation, in the future there will be new tools and procedures being used in the field. The tech industry is volatile due to it's wide range of applications, therefore it is important for engineers to be able to evaluate the ethics of the technologies being depended upon by the field.

This is very relevant to society at large as discussed in the ethical section, there are several issues that arise from this technology. The issues presented in this essay need to be more broadly discussed to encourage more positive adoptions of the technology to reduce the long term damage that is being currently done to the environment, the tech industry and the general public.

At a more personal level, this research has made me realize the wide impact this technology is having. It has made me think about how I apply the usage of GenAI. I mainly use it as a glorified documentation summarizer, but even then the research and understanding what it does makes my trust in it's correctness falter. I feel that there are ways to ensure there are guardrails that can be enforced such as code reviews, more statical analysis of the code with the tool and human made unit testing. However, at the end of the day, all of these practices are already best practices, so it really makes me wonder whether it is worth it to remove the

space for someone else to grow, damage the environment and displace people I will never meet. If I need to learn this tool for survival I will do so², but I now know how the tool works, the limitations and that while there is no clear accountability chain, for me to practice the virtues of honesty and fairness I should strive to be a person who knows the origins of my privileges and use them for bettering the world. While this is easy to say it is hard to practice, this field as mentioned before, is very volatile, so to be a good practitioner in the field, I need to understand not only how things work but how they affect those around me.

²As an F-1 student, my stay in this country depends on me finding work in the field I am pursuing, and given this is an industry trend, specially among large companies, I don't have the luxury of selection.

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